

JULIA ZIEMBA

QA Automation Engineer / SDET /
Software & AI Automation

juliamariaziemba@gmail.com

Bielsko-Biała

TECHNICAL SKILLS

Programming: Python · C++ · Bash/
zsh · C · JavaScript/TypeScript

QA: API Testing · Test Automation ·
Regression · Functional · Data
Testing

Automation: pytest · Playwright ·
Postman · Shell Scripting

Data: SQL · PostgreSQL · MySQL · S3
· Hadoop · Airflow · Jupyter

Tools: Git · Postman · Jira · pytest ·
Linux

AI: Claude · AI Agents · MCP ·
Process Automation

Familiar with: Node.js · Next.js ·
Docker · Kubernetes · Jenkins

EDUCATION

**Computer Science
(Engineering Degree)**

2019 – 2023

Uniwersytet im. Adama Mickiewicza
w Poznaniu

**Math-Physics-IT Profiled
Highschool**

2016–2019

III Liceum Ogólnokształcące im. S.
Żeromskiego w Bielsku-Białej

LANGUAGES

Polish (Native)

English (Bilingual Proficiency)

PROFILE

QA Engineer with 4+ years of experience across software testing, API automation, Data QA and AI-powered process automation. Strong programming background in Python, C++ and shell scripting, with experience building automated tests and internal tools from scratch. Particularly interested in automation and creating reliable, intuitive tools that make complex technical processes accessible to other users.

WORK EXPERIENCE

AI Automation / AI Enabling Team
Comscore (2026)

- Developed AI-powered agents designed to automate repetitive tasks for internal teams.
- Built an AI-powered QA assistant using **Anthropic Claude** that:
 - read Jira tickets and comments,
 - searched relevant Wiki documentation,
 - analysed requirements,
 - generated structured QA Reviews,
 - identified testing scenarios,
 - created Zephyr test cases with reproduction steps,
 - generated relevant Wiki documentation.
- Co-developed an extended version of the solution with my manager using an **internal MCP**, adding capabilities such as:
 - source code analysis,
 - enhanced documentation analysis,
 - BDD scenario generation,
 - expanded QA Review functionality.
- Worked as part of an AI enabling team focused on identifying repetitive workflows and automating them with AI agents.
- Developed workflow automation using Microsoft Copilot and Power Automate where required by the team's technical constraints.

Data QA Engineer

Comscore (2025 – 2026)

- Performed data quality testing across different data pipelines and processes.
- Created data tests and analyses using Python and Jupyter Notebook.
- Retrieved and analysed data from Amazon S3 and Hadoop.
- Worked with Airflow to execute and validate data-related processes.
- Helped define and expand regression testing scope for data.
- Supported multiple teams as a Data QA specialist.
- Automated repetitive data testing processes.

QA Engineer / API & Test Automation

Comscore (2024 - 2025)

- Transitioned into QA with little prior professional testing experience and rapidly developed a strong foundation in software testing principles.
- Performed API and functional testing.
- Created automated tests using Python, shell scripting and Postman.
- Designed and executed regression tests.
- Supported QA across multiple teams.
- Provided UI testing support when dedicated UI testers were unavailable.
- Used programming skills to automate previously manual testing tasks and processes.

IT Technical Product Support

Shareablee Inc./Comscore (2022 - 2024)

- Provided technical support and troubleshooting for the company's web application.
- Worked primarily on frontend-related issues, implementing smaller fixes and improvements that did not require involvement from the main development team.
- Worked with technologies including Django and TypeScript in the context of maintaining and troubleshooting the application.
- Investigated reported issues, identified their causes and implemented or coordinated appropriate fixes.
- Created technical reports and data extracts for other departments, including reports based on Grafana data.
- Worked across technical and non-technical teams to resolve issues and fulfil ad-hoc technical requirements.

SELECTED PROFESSIONAL PROJECTS

AI-Powered UI Testing with Claude, MCP & Playwright

Claude · MCP · Playwright · Python · Allure

Co-developed an automated UI testing system that uses Claude through an MCP integration to analyse Jira bug tickets and autonomously create and execute reproducible browser tests.

- Generates step-by-step reproduction instructions and corresponding Playwright test code from a Jira bug ticket.
- Executes browser interactions, captures screenshots/snapshots and test evidence, and generates Allure reports.
- Maintains a persistent website knowledge base (knowledge.md) describing navigation and UI interactions; the system can update this knowledge based on newly discovered information.
- Tracks token usage and execution time for individual scenarios.
- Optimised the architecture to generate and persist reusable Playwright code instead of invoking the LLM for every browser interaction.
- Reduced initial execution time from ~32 min to ~20 min and initial token usage from ~700k to ~600k tokens.
- Enabled subsequent executions of generated tests to run in ~2 min with no additional LLM token usage, making the solution particularly suitable for regression testing.
- Refactored and maintained the project structure to improve maintainability and execution performance.

Web UI for Automated Data Testing

HTML · CSS · JavaScript · Python · pytest · shell scripting · AWS

Designed and built a local web application from scratch to allow non-technical team members to run automated data tests without needing to understand Python or command-line tools.

Key functionality included:

- selecting the target environment and test suite,
- automatically selecting the appropriate AWS profile,
- automatically generating required data such as S3 dates and paths,
- selecting different URL testing modes,
- automatically preparing pytest configuration,
- launching tests directly from the interface,
- displaying test results in a user-friendly format,
- displaying passed/failed test counts and pass percentage,
- presenting logs and highlighting failed test areas,
- downloading logs for manual investigation,
- exporting results as PNG for sharing and confirmation,
- AWS profile login and authentication status checks.

The application transformed a command-line/Python-based testing workflow into a simple interface suitable for users without a technical background.

AI-powered QA Review Agent

Built an AI agent that automated a significant part of the QA Review preparation process.

The agent:

1. read a Jira ticket and its comments,
2. searched relevant Wiki documentation,
3. analysed requirements,
4. generated a structured QA Review,
5. identified relevant testing scenarios,
6. created Zephyr test cases and reproduction steps,
7. generated corresponding Wiki documentation.

The solution was later expanded together with my manager to use an internal MCP and support source-code analysis and BDD scenario generation.

PERSONAL & ACADEMIC PROJECTS

TwisteOptimizer

Python, SQLite, JavaScript | Personal project

Offline-first desktop planner for a collectible game, built without frameworks or third-party dependencies; the HTTP server uses only the Python standard library.

Aggregates three upstream sources into a local SQLite database, including a cross-language record-matching pass that reconciles ~514 Japanese and English card names at ~99% exact match with the remainder resolved by scoped similarity comparison; unmatched records are surfaced for review rather than inferred. Implements three independent, user-selectable scoring models with fully configurable weights, and a team-optimisation engine evaluating combinations against per-scenario rules.

Tender (Plant Care MCP)

Python, MCP SDK, FastAPI, SQLite | Personal project

Model Context Protocol server with an accompanying web interface, providing a deterministic reasoning engine for plant care scheduling and diagnostics. Care intervals are computed from species baselines adjusted for seasonality, latitude-derived daylight, container volume and live meteorological data. Indoor humidity is derived from outdoor conditions using the Magnus-Tetens saturation vapour pressure equation, to differentiate between conditions with similar symptoms, but different remedies for each. The diagnostic engine ranks causes across symptom-match and evidence signals, returning a test for each hypothesis. A local intent router resolves the majority of queries without model invocation; response is displayed to the user. Fully functional even without a Claude API key.

ShelfCheck

Python, FastAPI, Steam Web API, SQLite | Personal project

Decision-support web application for purchase recommendations based on a user's own historical spending and usage data. Computes a weighted score across four components, calculating projected engagement from similarity-weighted comparison against the user's existing library. Tag similarity uses inverse document frequency weighting to suppress non-discriminating common terms (like 'Free to Play'), and community ratings use the Wilson score interval, so small favourable samples do not outrank large ones. All scoring is deterministic and completed before any model call; model use is limited to query resolution and natural-language output, both with deterministic fallbacks.

Baimo

Python, scikit-learn, statistical analysis | Personal project

Current ongoing personal project to practice Machine Learning. Local wellbeing application combining rule-based NLP, statistical analysis and a supervised classifier. Implements a three-axis lexicon scorer handling negation, intensity modifiers, contrastive clauses and multi-word phrases.

MineCppraft

Unreal Engine / C++ | Personal project

Development of a Minecraft-inspired game using Unreal Engine and C++.

Microshell

C

First-year university project – implementation of a simple microshell in C, imitating basic zsh functionality.

Persistent Homology

Python

University project applying persistent homology using Python.

News Website Mockup

HTML / CSS

University project involving the visual design and implementation of a news website.